1. **A company has been allocated a Class C network 192.168.5.0/24. The physical networ should be divided into 5 subnets, which will be interconnected by routers. Class C custom subnets need to be designed. Derive the subnets that would meet the requirements shown outlining the;**

a.) **Network address (network ID)**

b.) **First and last usable IP addresses**

c.) **Broadcast address (broadcast ID)**

d.) **Hosts per subnet**

2^n = no. of subnets n=3 because 2^3 =8. This is the smallest power of greater than 5.

27-3=24. 24 is the new CIDR notation.

Subnet mask was, 255.255.255.0 /24. The subnet has now changed to 255.255.255.224

32-27=5 bits – total nuber of host assresses

2^5 =32 is number of host addresses

32-2= 30 Usable Host addresses per subnet

30 Usable Host addresses per subnet

To find the increment, 256-224=32

The subnets will be created by adding the increment to the base network address starting from 0.

i. **First Subnet**

192.168.5.0 /27

Network Address: 192.168.5.0

First Usable IP Address: 192.168.5.1

Last Usable IP Address: 192.168.5.30

Broadcast Address: 192.168.5.31

Hosts: 30

ii. **Second Subnet**

192.168.5.32 /27

Network Address: 192.168.5.32

First Usable IP Address: 192.168.5.33

Last Usable IP Adress: 192.168.5.62

Broadcast Address: 192.168.5.63

Hosts: 30

iii. **Third Subnet**

192.168.5.64 /27

Network Address: 192.168.5.96

First Usable IP Address: 192.168.5.97

Last Usable IP Address: 192.168.5.126

Broadcast Address: 192.168.5.95

Hosts: 30

iv. **Fourth Subnet**

192.168.5.96 /27

Network Address: !92.168.5.96

First Usable IP Address: 192.168.5.97

Last Usable IP Address: 192.168.5.126

Broadcast Address: 192.168.5.127
Hosts: 30

v. **Fifth Subnet**
192.168.5.128 /27
Network Address: 192.168.5.128
First Usable IP Address: 192.168.5.129
Last Usable IP Address: 192.168.5.158
Broadcast Address: 192.168.5.159
Hosts: 30

2. For each of the following IP addresses, carry out fixed length subnetting (FLSM) by applying the given subnet mask and utilizing additional masking bits borrowed from the default subnet mask. Determine the number of possible subnets per given network ID but only state the first three subnets, where possible. Consider the zero subnet in each of the below network IDs as your first possible subnet.
Additionally, in every subnet, determine the; number of hosts, network address/ID, first and last usable IP, and the broadcast address.

a) 192.168.10.0 /25

Subnet Mask: 255.255.2555.128/25
2^1=2
Hosts by Subnet: /25 32-25=7 bits for hosts
No. of hosts will be 2^7 =128
Usable hosts: 128-2=126

First Subnet
192.168.10.0/25
Network Address: 192.10. 0
First Usable IP Address: 192.168.10.1
Last Usable Address: 192.168.10.126
Broadcast Address: 192.168.10.127

Second Subnet
192.168.10.128/25
Network Address: 192.168.10.128
First Usable IP Address: 192.168.10.129
Last Usable IP Address: 192.168.10.254
Broadcast IP Address: 192.169.10.255

b) 192.168.10.0/28

255.255.2555.240 /28
No. of subnets 2^4 =16
32-28=4 bits for hosts
Usable hosts will be 16-2=14
Subnet 1:
Network Address: 192.168.10.0
First usable IP Address: 192.168.10.1
Last Usable IP Address: 192.168.10.14
Broadcast Address: 192.168.10.15

Subnet 2:
Network Address: 192.168.10.16
First Usable IP Address: 192.168.10.17
Last Usable IP Address: 192.168.10.30
Broadcast Address: 192.168.10.31
Subnet 3:
Network Address: 192.169.10.32
First Usable IP Address: 192.168.10.33
Last Usable IP Address: 192.168.10.46
Broadcast Address: 192.168.10.47

c) 10.0.0.0/30

255.255.255.252 /30
2^6 =64
/30 32-30= 2 bits per host
No.of hosts will 2^2 =4
Usable hosts 4-2=2
Subnet 1:
10.0.0.0/30
Network Address: 10.0.0.0
First Usable IP Address: 10.0.0.1
Last Usable IP Address: 10.0.0.2
Broadcast Address: 10.0.0.3
Subnet 2:
10.0.0.4/30
Network Address: 10.0.0.4
First Usable IP Address: 10.0.0.5
Last Usable IP Address: 10.0.0.6
Broadcast Address: 10.0.0.7
Subnet 3:
10.0.0.8/30
Network Address: 10.0.0.8
First Usable IP Address: 10.0.0.9
Last Usable IP Address: 10.0.0.10
Broadcast Address: 10.0.0.11

d.)10.0.0.0/16

Subnet Mask is 255.255.0.0 /16

No bits will be borrowed so there will be only one subnet

2^0 =1

No. of hosts per subnet:

32-16=16 bits for the hosts

No. of hosts 2^{16} =65536

Usable hosts will be 65536-2=65534

Subnet:
10.0.0.0/16

Network Address: 10.0.0.0

First Usable IP Address: 10.0.0.1

Last Usable IP Address: 10.0.255.254

Broadcast Address: 10.0.255.255

e.) 172.16.0.0/30
255.255.255.252 /30
No. of subnets will be 2^6 =64
No. of hosts per subnet 32-30=2 bits for hosts
2^2 =4
4-2=2
Subnet 1:
172.16.0.0 /30
Network Address: 172.16.0.0
First Usable IP Address: 172.16.0.2
Broadcast Address: 172.16.0.3
Subnet 2:
172.16.0.4 /30
Network Address: 172.16.0.4
First Usable IP Address: 172.16.0.5
Last Usable IP Address: 172.16.0.6
Broadcast Address: 172.16.0.7
Subnet 3:
172.16.0.8 /30
Network Address: 172.16.0.8
First Usable IP Address: 172.16.0.9
Last Usable IP Address: 172.16.0.11

f.)172.16.0.0/17

Subnet is 255.255.128.0 /17
No. of subnets is 2 2^1 =2
No. of hosts per subnet will be 32-17= 15 bits for the hosts
No. of hosts will be 2^{15} =32768
Usable hosts will be 32768-2=32766
Subnet 1:
172.16.0.0 /17

Network Address: 172.16.0.0
First Usable IP Address: 172.16.0.1
Last Usable IP Address: 172.16.127.254
Broadcast Address: 172.6.127.255
Subnet 2
172.16.128.0 /17
Network Address: 172.16.128.0
First Usable IP Address: 172.16.255.254
Broadcast Address: 172.16.255.255